

Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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Hospital de
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Expert Clinical Proposal: Expected Healthy Vessel Modeling

To address the intrinsic limitations of the rigid 10 mm reference window, clinical experts at Hospital de la Santa Creu i Sant Pau proposed a paradigm shift for future iterations of the quantification algorithm. The core problem with the current local reference method is "reference contamination": if the adjacent 10 mm segments are also affected by diffuse disease, physiological tapering, bifurcations, or remodeling, the baseline area is compromised, leading to inaccurate stenosis percentages.

To solve this, the experts propose moving from a local relative calculation to the construction of an **Expected Healthy Vessel Model**. This approach estimates the theoretical anatomical caliber the vessel would have at any given point if no atherosclerotic plaque were present. The proposed methodology involves four sequential steps:

1. **Comprehensive Feature Extraction:** For every point along the extracted centerline, the algorithm should compute not only the luminal area but also the effective diameter, distance from the ostium, local curvature, and proximity to bifurcations.
2. **Exclusion of Unreliable Zones:** Before calculating any baseline, the system must actively exclude non-reliable segments from the mathematical fitting process. This includes regions with evident stenosis, carinas/bifurcations, the ostium, areas with severe calcification artifacts, or sudden non-physiological diameter drops.
3. **Robust Reference Curve Fitting:** Instead of relying on adjacent 10 mm samples, the algorithm should fit a continuous, smooth reference curve along the entire valid length of the vessel. Techniques such as robust spline regression, robust LOESS, or hierarchical Bayesian modeling can mathematically reconstruct the natural physiological tapering from proximal to distal ends. The stenosis calculation would then be updated to:
$$\% \text{ Area Stenosis} = 100 \times (1 - \frac{\text{Observed Minimal Area}}{\text{Expected Reference Area}})$$
4. **Integration of Population-Based Priors:** Finally, to maximize robustness, the reference curve should be trained and constrained using population data from healthy coronary vessels. The model would adjust the expected caliber based on patient-specific and anatomical variables, such as the specific coronary territory (LAD, LCX, RCA), SCCT segment nomenclature, coronary dominance, patient sex and body size, and myocardial volume.

Implementing this advanced modeling strategy would effectively eliminate the primary sources of quantification error, ensuring that diffuse disease, complex bifurcations, and naturally small distal vessels no longer distort the healthy reference baseline.